Exercise 7

Draw the TCP/IP Model: Application Layer → Transport Layer → Internet Layer → Link Layer

Provide at least two protocol examples for each of the four TCP/IP layers.:

1. Application Layer

- HTTP (HyperText Transfer Protocol) → web browsing
- FTP (File Transfer Protocol) → file transfer
- SMTP (Simple Mail Transfer Protocol) → email
- DNS (Domain Name System) → resolving domain names

2. Transport Layer

- TCP (Transmission Control Protocol) → reliable, connection-oriented
- UDP (User Datagram Protocol) → unreliable, connectionless
- DCCP (Datagram Congestion Control Protocol)
- SCTP (Stream Control Transmission Protocol)

3. Internet Layer

- IP (Internet Protocol) → addressing and routing
- ICMP (Internet Control Message Protocol) → error reporting, diagnostics (e.g., ping)
- IGMP (Internet Group Management Protocol) → multicast management

4. Network Access / Link Layer

- Ethernet → wired LAN protocol
- Wi-Fi (IEEE 802.11) → wireless LAN protocol
- PPP (Point-to-Point Protocol) → serial connections
- ARP (Address Resolution Protocol) → maps IP addresses to MAC addresses

Wireshark notes:

- When intiating packages capture we can set a default filter, for instance tcp this way our captured packages will only be the one that uses tcp protocol.
- We can use another filter which is display filter this filters all the captured packages according to our needs one of those display filters is ip.address == 192.123.11 this is a random ip and probably an invalid one
- eth.address == 000:ff:2332 this helps you filter according to mac address
- we have logical operators like a programming language this can be (not, and, or) we literally write them we don't use signs for them in the wireshark filter

Identify the various protocols necessary for contacting the web ser ver up to displaying the website in the web browser. Write/Copy these in the order they appear: TLS, TCP, DNS, MDNS, UDP, QUIC